Genomic Insights into Extreme Salinity Tolerance in *Salicornia*

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Summary

- Chromosome-scale, annotated genome assemblies were generated for six *Salicornia* species.
- Four distinct subgenomes were identified within the *Salicornia* genus.
- Comparative genomics revealed lineage-specific expansions of the Cupin_1 and CHS gene families
- Global resequencing panel established for both genetic research and breeding applications.
- A database of functionally validated salt stress genes was curated via large-scale literature review using LLM agents.

Introduction

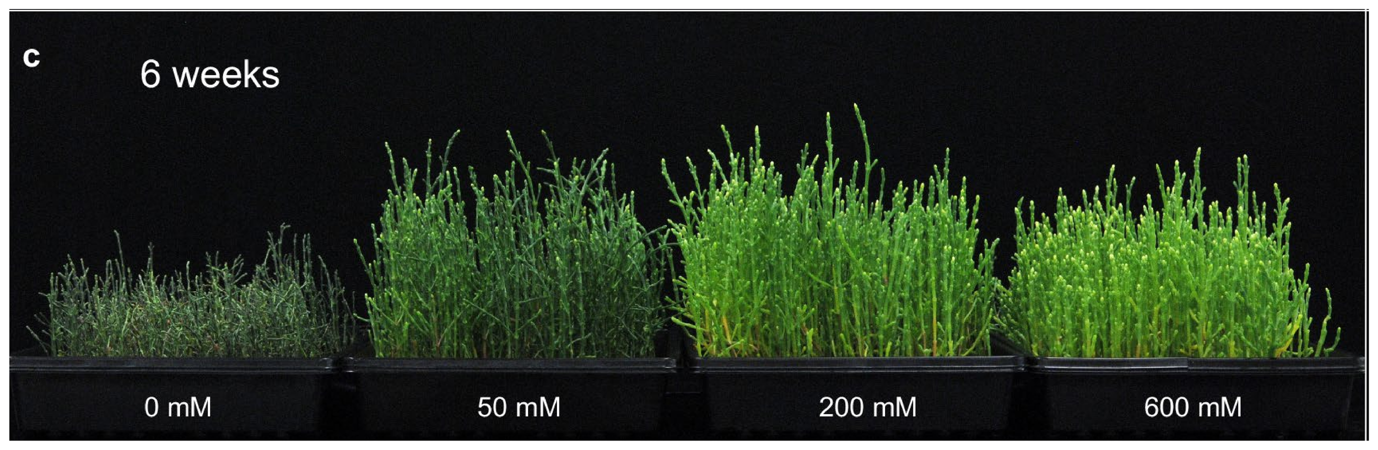
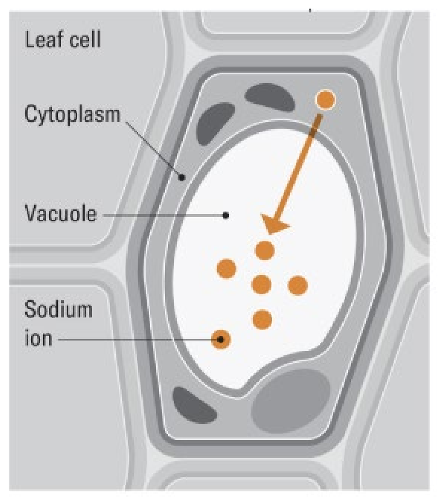


Figure 1. Plants were grown for 6 weeks under NaCl concentrations of 0, 50, 200, 600 mM.
a: Plants 6 weeks after treatment. b: Length of individual plants 6 weeks after treatment. (From Salazar et al 2024)



From Jen Christiansen and Tim Flowers

Saltwater-based agriculture is a sustainable solution to address escalating water scarcity and the growing demand for irrigation water for cropping. Halophytes such as ***Salicornia***, a genus in the Amaranthaceae family, is valued for its edible succulent stems and oil-rich seeds, yet its genetic improvement has been constrained by limited genomic resources.

Salicornia plants grow optimally at around 200 mM NaCl, and still grow and complete their life cycle at 600 mM NaCl(Fig 1); these conditions are lethal to most plants. Remarkably, salt promotes the growth of *Salicornia* spp and a mechanism of tolerance is to sequester the sodium into the vacuole of shoots - referred to as tissue ion tolerance.

Material - Seeds Collection

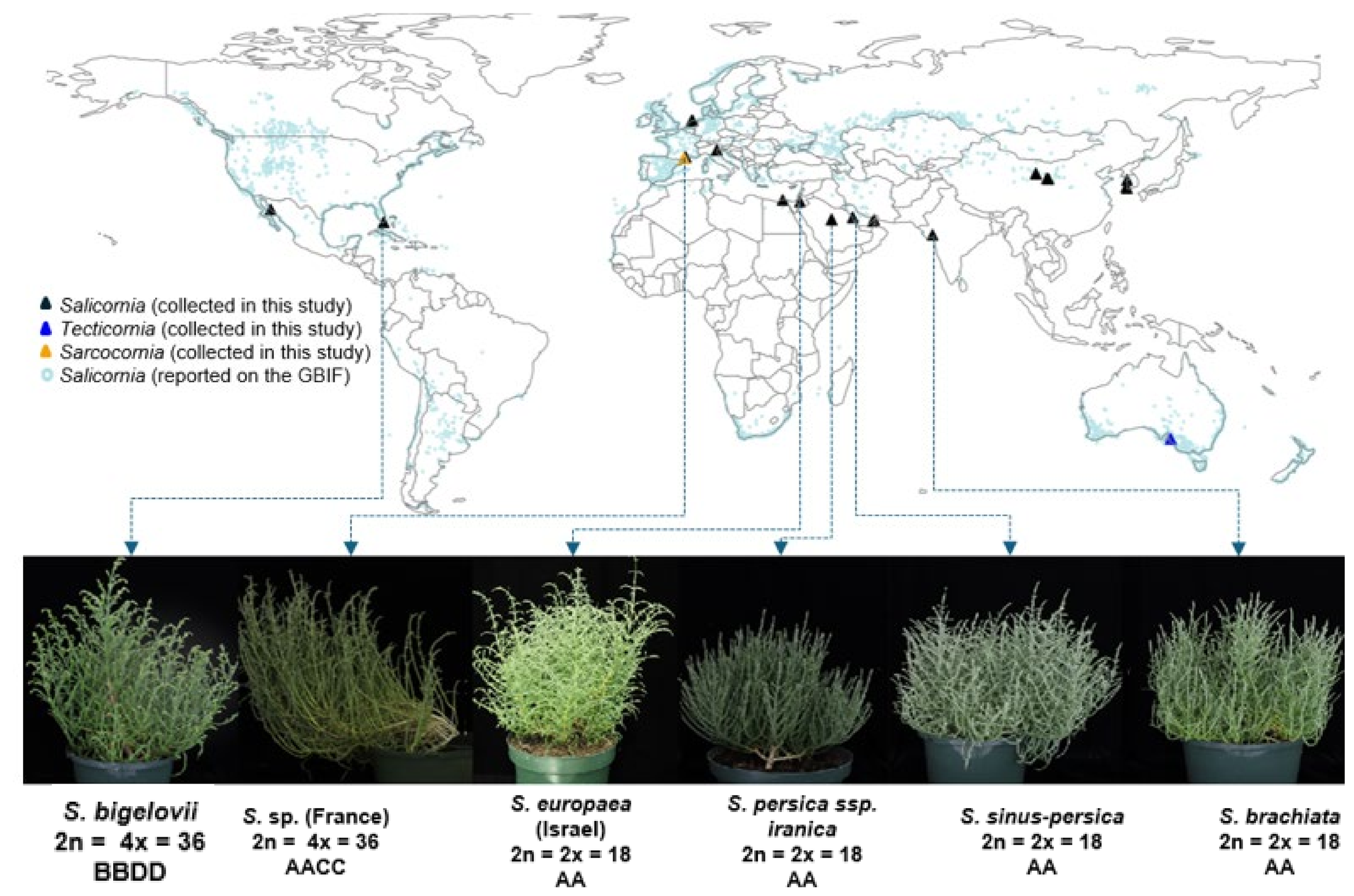


Figure 2. Geographic distribution of Salicornioideae and morphological characteristics within the *Salicornia* genus
Geographic distribution of *S. bigelovii*, *S. europaea* (France), *S. europaea* (Israel) , *S. persica ssp. iranica*, *S. sinus-persica*, *S. brachiata* and their diverse morphological characteristics. Distribution data (light blue dots) from GBIF.org

Results

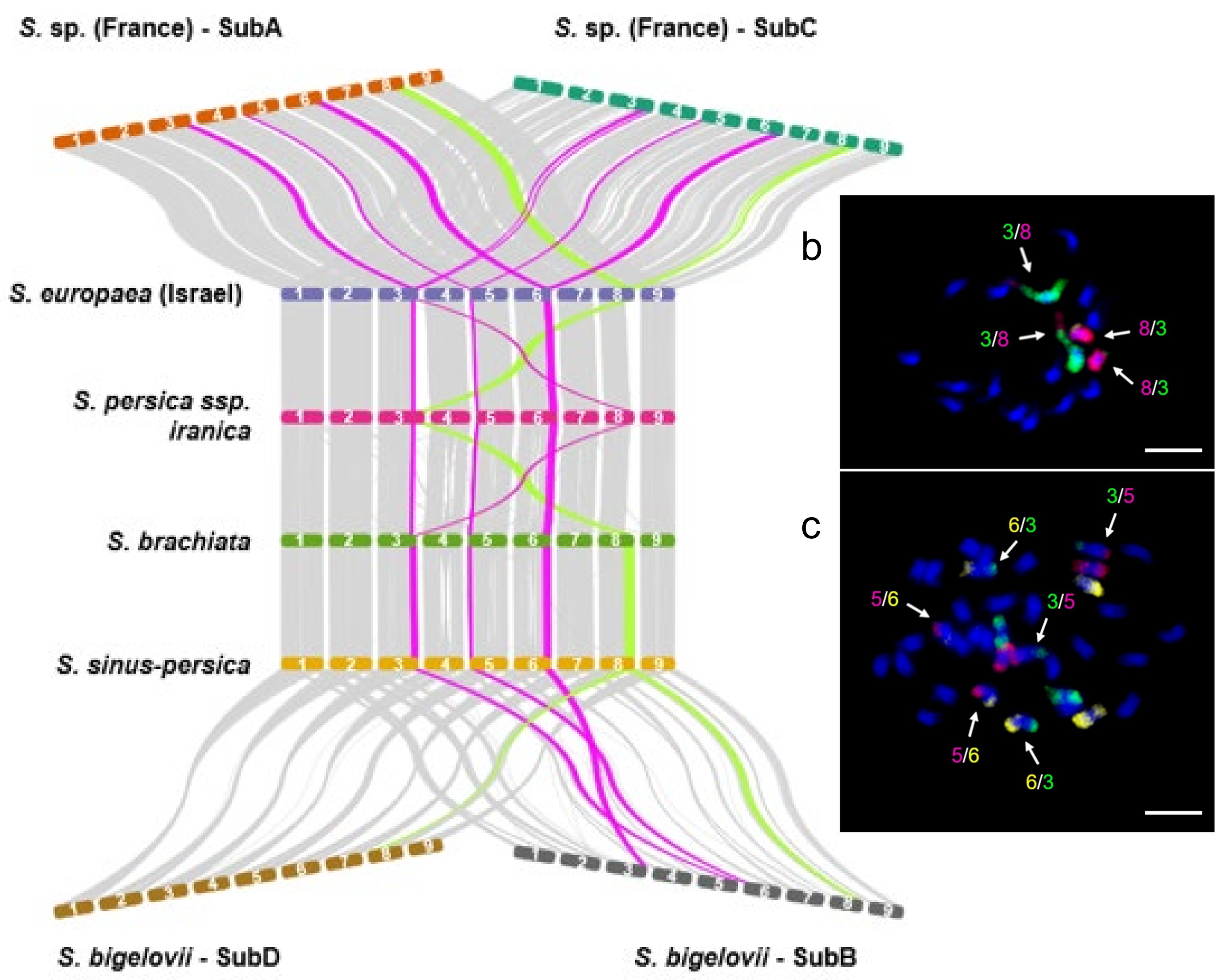


Figure 3. Intergenomic synteny among *Salicornia* lineages.
a: Synteny blocks are connected in gray. The large reciprocal translocations in chromosome three with chromosome eight in *S. persica ssp. iranica* are highlighted in light green. Other Large reciprocal translocations are highlighted in pink including. b: Reciprocal translocations revealed by oligo-FISH (*S. europaea* (Israel)) on mitotic metaphase chromosomes of *S. persica* ssp. *iranica* and c: *S. bigelovii*.

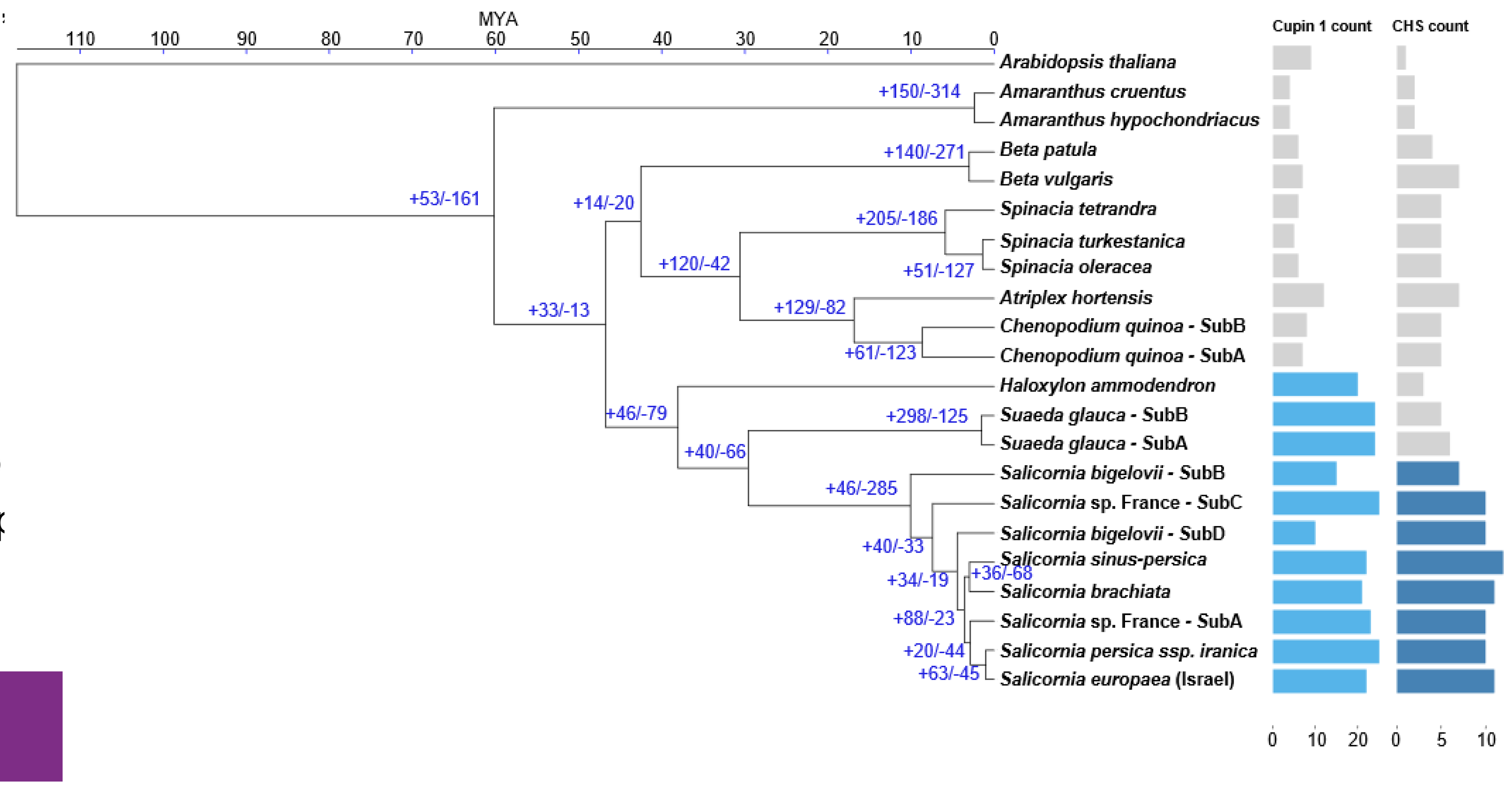


Figure 4. Orthogroup-based phylogeny, copy number variation, and gene family expansion in the Amaranthaceae family.
A time-calibrated phylogenetic tree built from single-copy ortholog protein sequences using maximum likelihood. The top axis shows divergence time (MYA). Blue numbers at internal nodes denote inferred gene-family expansions (+) and contractions (−) on the branch leading to each node ($p < 0.05$; CAFÉ). Right columns show per-genome gene counts for the Cupin_1; left) and chalcone synthase (CHS; right) families, with bar lengths proportional to copy number. Light blue: species with Cupin_1 family expansions; dark blue: species with CHS family expansions; grey: species without expansions.

Reference

Wang, Y., Adhikari, L., Leal, L. C., Kovalchuk, N., Čížková J, Beránková D, ... Tester, T. & Melino, V. (2025). From Wild Halophyte to Future Crop: Genomic Insights into *Salicornia*. (preprint, DOI: 10.21203/rs.3.rs-7061928/v1)
Salazar, O. R., Chen, K., Melino, V. J., Reddy, M. P., Hřibová, E., Čížková, J., ... & Schmöckel, S. M. (2024). SOS1 tonoplast neo-localization and the RGG protein SALTY are important in the extreme salinity tolerance of Salicornia bigelovii. Nature Communications, 15(1), 4279.

A Literature-Curated Salt Stress Gene Resource Built by Multi-LLM Consensus

PubMed corpus acquisition
PubMed
Salt-stress + plant query
122,617 abstracts

Topic filtering
KeyBERT + UMAP + HDBSCAN
Haiku 4.5
Noise: 55,644 abstracts
Retained for LLM extraction:
n = 45,639 abstracts

Noise rescue
Keyword-density filter on noise
55,644 → 52 candidates → sonnet + Dual-LLM gene extraction
+21 rows genes

Dual-LLM gene extraction
Sonnet 4.6 → 45,107 calls
Opus 4.7 → 44,800 calls
Agreement: 6,112 abstracts
Disagreement:1,403 abstracts
Opus adjudication: keep 1,748 | drop 669

Entity normalization
NHX7=SOS1
Alias clusters(gene-level): 7,938
Unique species(normalized): 808
Unique PMIDs with gene: 6,968

Database construction
GitHub-based Website

Full-text extension
Full text Download:
PMC OA 3,228 + KAUST 2,262
5,490/6,902 papers(79.5%)
Sonnet 4.6 → 5,482/5,490
Opus 4.7 → 5,444/5,490

Salt-stress gene table
9,849 rows
7,938 unique gene
808 species
6,968 supporting PMIDs

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